

EXPERIENCE LAB INTERNSHIP PROGRAMME

Week 7: Project Summary

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HealthConnect Clinic Experience Lab*Data Science Track — Model Testing, Error Analysis & Refinement*

Programme	AnalystLab Africa Experience Lab
Week	Week 7
Project	HealthConnect Clinic Experience Lab
Track	Data Science

1. What Was Planned to Test

The plan was to systematically test the Week 6 candidate model beyond aggregate metrics, conduct deeper error and segment analysis, verify whether its Week 6 improvement was meaningful, and complete a genuine cross-track testing activity with the Data Analytics track, testing a real hypothesis rather than simply exchanging findings.

2. What Was Actually Tested

- Reproduced and confirmed the Week 6 candidate model's performance as a starting baseline for this week's testing.
- Conducted error analysis profiling false positives and false negatives in more depth than Week 6.
- Tested model performance across booking lead time segments specifically, not just in aggregate.
- Checked for overfitting by comparing training and test performance.
- Reviewed whether existing features still carry signal within the weakest-performing segment.
- Independently validated a combined Data Analytics finding (booking lead time $\times$ reminder status), then tested whether encoding it as an interaction feature improved the model.

3. Most Important Testing Results

Segment-level testing revealed that the model's overall 66% recall is concentrated almost entirely in long-lead-time appointments (91% recall) while short-lead-time appointments are barely predicted at all (6–16% recall). This imbalance was invisible in Week 6's aggregate metrics and is the most significant finding of the week.

4. Issues or Weaknesses Identified

- Severe recall imbalance across booking lead time segments.
- The model appears to under-use reminder status and prior history signals specifically when lead time is short, despite those signals carrying real information (confirmed via within-segment analysis).
- Logistic Regression's linear structure may itself be the limiting factor, not the available features.

5. Improvements/Refinements Made

An interaction feature combining booking lead time category and reminder status was engineered and tested, based on a validated finding shared by the Data Analytics track.

6. Retesting Results

The interaction feature produced negligible aggregate change (accuracy +0.002) and no meaningful change in segment-level recall (0-7 day recall unchanged at 6.1%). This was a validated negative result: the hypothesis was

reasonable and worth testing, but the evidence did not support adopting the change. The original two-feature representation was retained.

7. Track(s) Collaborated With

Data Analytics (primary, continuing from Week 6).

8. What Was Tested Collaboratively

The Data Analytics track's validated finding that the booking-lead-time no-show pattern holds even when reminder status is considered, and within every lead-time group, a reminder is associated with a lower no-show rate. This was independently reproduced exactly, then tested as a candidate model refinement.

9. What Changed as a Result

No feature change was ultimately adopted, but a plausible fix was ruled out with evidence, narrowing the search for what will actually resolve the segment imbalance, most likely a non-linear model or a segment-specific decision threshold, both flagged for Week 8.

10. Key Findings or Validation Outcomes

- No evidence of overfitting (training and test performance nearly identical).
- Features remain informative even in the weakest-performing segment; the limitation is in how the model uses them, not the data itself.
- A cross-track-validated interaction feature does not resolve the model's core weakness.

11. Major Challenges Encountered

The main challenge was analytical discipline: a small aggregate improvement (+0.002 accuracy) could have been reported as a success, but the segment-level data made clear that the actual problem was unresolved. Reporting the negative result honestly, rather than the flattering aggregate number, was the more useful outcome.

12. Important Decisions Made

Chose to retain the original two-feature representation over the interaction feature, since the interaction added complexity without measurable benefit to the segment that actually needed improvement.

13. Remaining Limitations

See the full list in the main notebook (Task 21): principally, the unresolved short-lead-time recall gap, no hyperparameter tuning attempted yet, and reliance on a single train/test split rather than cross-validation.

14. Remaining Issues or Dependencies

A response is owed to the Data Analytics collaborator with this week's test outcome. A technical handoff to ML Engineering regarding the segment limitation remains to be finalised.

15. Contribution to the Overall HealthConnect Project

This week's testing surfaced a materially important, previously invisible weakness in the candidate model, and ruled out one plausible fix with rigorous evidence rather than assumption, giving the project a more honest and useful foundation heading into final integration.

16. What Must Be Completed Before Week 8

Test a properly tuned Random Forest and a segment-specific classification threshold to address the short-lead-time recall gap; run patient-aware cross-validation to confirm these findings are stable